Paper Title: Symmetric Stigmergy with Foundation-Model Agents: Pheromone-Field Coordination for Decentralized Artifact Refinement

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Submission to: Special Issue "When Foundation Models Meet Multi-Agent Systems"

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1. What is the main claim of the paper? Why is this an important contribution to the autonomous agents and multi-agent systems literature?

Main claim. Stigmergy works not only with simple reactive agents but with reasoning foundation-model agents, and a concrete system instantiates Parunak's symmetric-stigmergic schema (MABS25) in this setting. In the regime where strict-greedy hierarchical control collapses into a rejection loop, decentralized symmetric-stigmergic coordination solves problems that hierarchical and conversation-based coordination cannot; on the smallest instances, where that loop does not trigger, the advantage narrows.

Why this matters for MAS. The symmetric-stigmergic schema is recent, and its value depends on whether it admits agents more capable than ants. This paper supplies that demonstration: it maps a full foundation-model artifact-refinement loop onto the schema's primitives, including the environment-to-agent arrow (ASU) that classic asymmetric stigmergy lacks, realized as the environment escalating agent sampling and model capability when the sensed field stalls. The contribution is bidirectional in the sense the special issue asks for: foundation models supply the open-ended action proposals that classic stigmergy could not enumerate, and the symmetric schema supplies the objective, pressure-based criterion for combining those proposals and for allocating capability. The empirical core is the single-application comparison the editor identified as attractive: five coordination mechanisms on one application, under identical models, prompts, problem instances, and budget.

2. What is the evidence you provide to support your claim? Be precise.

Empirical evidence. A regenerated grid of 1350 trials (5 coordination strategies $\times$ {1, 2, 4} agents $\times$ {easy, medium, hard} difficulty $\times$ 30 trials) compares pheromone-field, conversation, hierarchical, sequential, and random coordination on meeting room scheduling. All strategies use identical models (qwen2.5 0.5b/1.5b/3b), identical prompts and parsing, the same deterministically seeded problem instances, the same 50-tick budget, and one authoritative pressure metric. The results show pheromone-field coordination leading aggregate solve rate, and being the only strategy to reach non-zero solve rates on medium and hard instances, where the baselines collapse through the rejection loop. A nine-arm ablation, including a `no_propagation` arm, isolates the contribution of evaporation, inhibition, the one-hop ESP propagation, and few-shot examples. Solve rates carry Wilson confidence intervals, effect sizes are reported per comparison, and ratio claims (final pressure, convergence speed) are reported with their unsolved-trials and small-sample caveats. The grid was regenerated after a code audit corrected a baseline system-prompt confound and a cross-strategy metric mismatch; the complete results with confidence intervals and figures accompany the manuscript.

Theoretical evidence. One lemma establishes that aligned local descent terminates, in a number of ticks bounded by the initial pressure rather than by the size of the state or action space. The scheduling domain's per-region pressure is separable ($\epsilon = 0$), so the lemma's alignment condition holds; an appendix reports the empirical check.

3. What papers by other authors make the most closely related contributions, and how is your paper related to them?

Symmetric stigmergy and the pheromone formalism. Parunak (2025, *Insects and Agents*, MABS at AAMAS) introduces the symmetric-stigmergic schema this paper instantiates; we realize each primitive, including the *ASU* arrow, with foundation-model agents. Parunak (2011, *Between Agents and Mean Fields*, MABS at AAMAS, 106–117) establishes the pheromone-field continuum and the aggregate/propagate/evaporate vocabulary the method adopts. Parunak (1997, *Annals of Operations Research*, 75, 69–101) gives the engineering-principles account our deposition-and-evaporation core inherits.

Environment-mediated coordination for foundation-model agents. Agashe et al. (2025, NAACL Findings) find language-model agents strongest when coordination rests on environmental state and weakest at theory-of-mind reasoning, directly supporting this design; Park et al. (2023, UIST) coordinate through environmental co-presence, where we instead make the environment-to-agent arrow explicit; Riedl (2025) measures emergent coordination under group-level feedback with no direct messaging.

Stigmergy lineage and scaling. Dorigo and Stützle (2004, *Ant Colony Optimization*, MIT Press) coordinate over enumerated spaces, the restriction foundation models lift; Mamei and Zambonelli (2009, *ACM TOSEM*, 18(4)) provide field-based middleware of which the sensed field is an instance; Xu et al. (2022, *IEEE TNNLS*, 33(9)) formalize digital pheromones for learning agents; Kim et al. (2025) show coordination overhead can become a net cost, motivating the $O(1)$-overhead design; Savarimuthu, Ranathunga, and Cranefield (2024, COINE at AAMAS) treat normative reasoning, complementary to the pressure function's continuous behavioral signal.

Relationship summary. The paper forms one lineage from digital pheromones to language-model agents. Where ant colony optimization coordinates over enumerated solutions, foundation-model actors propose open-ended patches; where blackboard and co-presence systems leave the environment-to-agent influence implicit, this system implements it as escalation; and where conversation-based frameworks combine outputs through dialogue, the sensed field adjudicates them objectively.

4. Have you published parts of your paper before?

An earlier version was posted to arXiv (January 2026) as a preprint. This is the first submission to a peer-reviewed venue; no prior conference publication exists. The present manuscript differs substantially from the preprint: the contribution is reframed as an instantiation of Parunak's symmetric-stigmergic schema with its exact notation; pheromone propagation over the domain's temporal structure is implemented and ablated; the empirical results are fully regenerated following a code audit that corrected a baseline confound and a metric mismatch; and the manuscript is reduced to 53 pages with claims scoped to the regime in which they hold. The bidirectional contribution stated under Question 1 addresses the special-issue theme directly.